

Robust Pilot-Tone-Based Timing Recovery Scheme for Coherent Digital Subcarrier Multiplexing Systems

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Abstract A robust, high-precision, and fast-converging frequency-domain pilot-aided timing recovery scheme for DSCM systems is proposed. By dividing two pilot tones, the scheme achieves FO/PN immunity. Experiments show strong robustness to roll-off and SFO, ensuring stable operation up to 280-ppm SFO. ©2026 The Author(s)

Introduction

Driven by the massive expansion of applications such as large language models, cloud computing, and the Internet of Things, coherent optical interconnect technology is evolving towards data rates of 1.6 Tb/s [1–3]. Digital subcarrier multiplexing (DSCM) is considered a more competitive candidate for future high-capacity optical interconnects than traditional single-carrier schemes, owing to its superior tolerance to equalization-enhanced phase noise and fiber nonlinearities [4–6]. Timing recovery (TR) is a critical technology that ensures the stable operation of high-speed coherent optical communication systems.

Existing TR schemes for DSCM typically operate under oversampling conditions. The computational complexity of performing timing error detection (TED) for each subcarrier individually scales linearly with the number of subcarriers [7, 8]. To reduce this complexity, Zhang et al. [9] and Li et al. [10] proposed schemes that utilize only a single subcarrier for TED. In Zhang's approach, the detection result is used for interpolating the entire DSCM signal, whereas Li's method applies the detection result to interpolate each subcarrier individually. These TR schemes are highly sensitive to the roll-off factor. Zhao et al. proposed a time-domain pilot-based scheme for DSCM systems, using inserted training sequences to extract timing information [11]. Although this method provides high robustness to roll-off factors and polarization crosstalk, it introduces additional transmission overhead, thereby reducing the net data rate. Conversely, Wang et al. proposed a frequency-domain pilot-based TR scheme for single-carrier self-coherent systems [12]. However, this scheme is inherently sensitive to frequency offset (FO) and phase noise (PN) as it relies on a self-coherent detection architecture.

To address these limitations, this paper proposes a frequency-domain pilot-aided TR scheme for DSCM systems. The proposed scheme ex-

tracts timing information via a pilot-division operation, which effectively cancels out residual FO and PN. By leveraging this mechanism, the recovered timing information becomes immune to phase fluctuations, thereby enabling highly robust TR performance independent of carrier synchronization. Experimental results demonstrate that the proposed scheme achieves significantly faster convergence than the Gardner algorithm and maintains stable performance across an ultra-wide sampling frequency offset (SFO) range of up to 280 ppm, even under a low roll-off factor of 0.01. This work provides a high-precision and robust synchronization solution for future high-speed coherent interconnects.

Principle

Fig.1(a) shows the schematic spectrum of the polarization-division multiplexing (PDM) DSCM signal at the transmitter. Two pilots are inserted in the X polarization, and one pilot is inserted in the Y polarization. The transmitted signal is expressed as:

$$E_{\text{DSCM},X}(t) = S_{\text{DSCM},X}(t) + A_{\text{PtX},1}e^{j\omega_{\text{PtX},1}t} + A_{\text{PtX},2}e^{j\omega_{\text{PtX},2}t}, \quad (1)$$

$$E_{\text{DSCM},Y}(t) = S_{\text{DSCM},Y}(t) + A_{\text{PtY},1}e^{j\omega_{\text{PtY},1}t}$$

where $E_{\text{DSCM},X}(t)$ and $E_{\text{DSCM},Y}(t)$ are the transmitted signals in the X and Y polarizations, respectively; $S_{\text{DSCM},X}(t)$ and $S_{\text{DSCM},Y}(t)$ are the corresponding subcarrier multiplexed signals; and $A_{\text{PtX},1}$, $A_{\text{PtX},2}$, $A_{\text{PtY},1}$ are the amplitudes of the pilots in each polarization. Fig. 1(b) illustrates the overall DSP flow at the receiver incorporating the proposed pilot-aided TR scheme. First, polarization demultiplexing and carrier phase recovery (CPR) are performed based on the frequency-domain pilots. Then, for TR, the two pilots in the X polarization are extracted:

$$\begin{bmatrix} P_{\text{TE},1}[n] \\ P_{\text{TE},2}[n] \end{bmatrix} = \begin{bmatrix} e^{j[\omega_{\text{PtX},1}(nT_a + \tau) + \kappa[n]]} \\ e^{j[\omega_{\text{PtX},2}(nT_a + \tau) + \kappa[n]]} \end{bmatrix}, \quad (2)$$

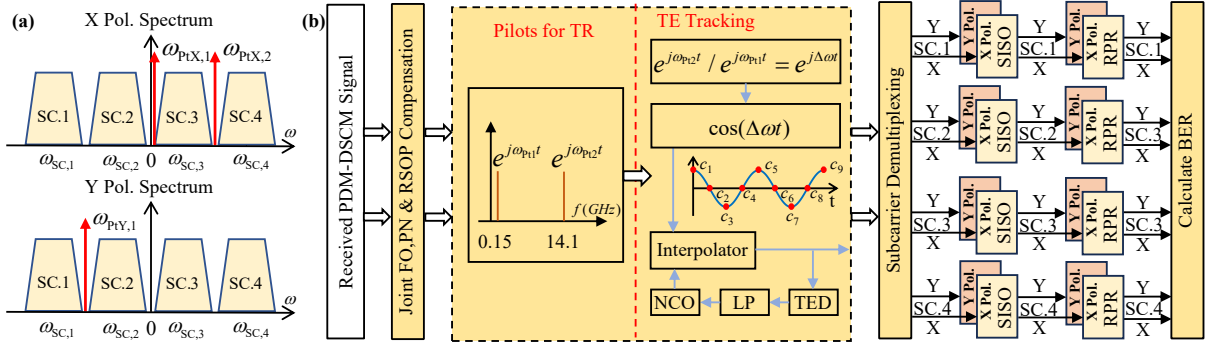


Fig. 1: (a) Schematic of the PDM-DSCM signal spectra at the transmitter, and (b) overall receiver DSP workflow incorporating the proposed pilot-aided TR. LP: loop filter. NCO: numerically controlled oscillator. RPR: residual phase recovery.

where T_a is the sampling period, τ is the sampling error, and $\kappa[n]$ is the residual phase noise introduced by the phase noise compensation module due to the presence of a timing error during phase noise compensation. The division of the two pilots eliminates the effect of the residual phase noise $\kappa[n]$, yielding a signal that contains only the timing error, which can be expressed as:

$$P_r[n] = \frac{P_{TE,2}[n]}{P_{TE,1}[n]} = e^{j\Delta\omega_{Pt}(nT_a + \tau)}. \quad (3)$$

The real part of $P_r[n]$ is a cosine signal:

$$P_c[n] = \cos[\Delta\omega_{Pt}(nT_a + \tau)], \quad (4)$$

where $\Delta\omega_{Pt} = \omega_{PtX,2} - \omega_{PtX,1}$. When $\frac{\Delta\omega_{Pt}}{2\pi}$ equals one quarter of the ADC sampling rate f_s , $P_c[n]$ exactly satisfy the requirements for TED calculation using the Gardner algorithm, as illustrated in Fig. 1(b). Here, interpolation is used instead of adjusting the ADC sampling rate. To enhance precision, cosine interpolation is applied to $P_c[n]$ within the TED loop. The derived timing information is then mapped to the DSCM signal for TR via conventional polynomial interpolation. Next, subcarrier demultiplexing is performed. A single-input single-output (SISO) equalizer based on the radius-directed equalization (RDE) algorithm compensates for linear impairments such as bandwidth limitation, followed by residual phase recovery (RPR) using the 16QAM VVPE algorithm. Finally, the bit error rate (BER) is calculated.

Simulation

The performance of different TR schemes is evaluated in a 48-GBaud PDM-16QAM transmission system. To evaluate the statistical stability of the proposed scheme, Fig. 2 illustrates the superimposed S-curves for 1,000 independent TED evaluations at an OSNR of 20 dB. As indicated by the heatmap, the S-curves exhibit a highly concentrated distribution with a clear sinusoidal profile. The narrow red region suggests that the timing error extraction is highly consistent, even in the presence of amplified spontaneous emission noise.

The corresponding timing jitter is calculated to be -48 dB, confirming the high estimation precision of the proposed scheme.

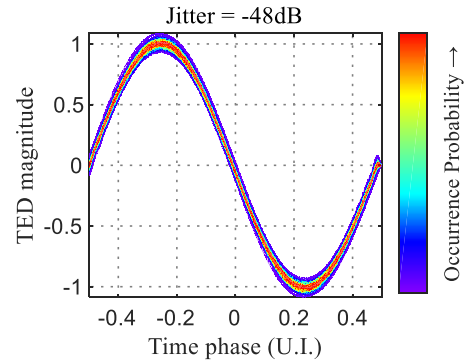


Fig. 2: S-curve and timing jitter performance at 20 dB OSNR.

Fig. 3 presents the convergence trajectories under a worst-case initial timing offset of 0.5 unit interval (U.I.). The proposed scheme achieves the fastest convergence, locking within approximately 2×10^3 symbols. In contrast, both the Gardner and Sign MM algorithms require about 2×10^4 symbols. Notably, under this large initial error, the Gardner algorithm fails to converge to the desired 0.5 U.I.. The proposed algorithm maintains high stability and precision after convergence, whereas the Gardner and Sign MM algorithms exhibit significant fluctuations. Under a 40-ppm SFO, the mean square error (MSE) of the proposed scheme is 2.6×10^{-5} , which is significantly lower than those of the Gardner (6.9×10^{-3}) and Sign MM (3.8×10^{-3}) algorithms.

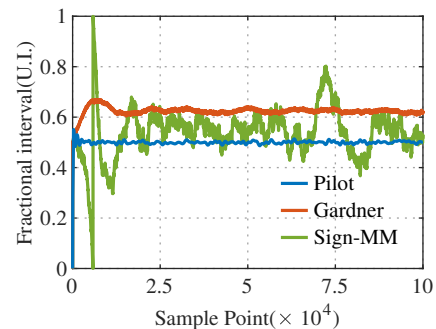


Fig. 3: Fractional interval convergence of three TR scheme.

EXPERIMENTAL DEMONSTRATION

The experimental setup and DSP flow for the proposed scheme are illustrated in Fig. 4. At the transmitter (Tx), DSCM signals with 4 subcarriers are generated, with symbol rates of 12 Gbaud for each subcarrier. Pulse shaping was achieved using a root-raised cosine filter. The guard band between subcarriers is 1 GHz. The pilot frequencies used for TR are 0.15 GHz and 14.1 GHz, respectively. The double-sided bandwidth of the DSCM signal is 55.8 GHz. The signal was converted into an analog electrical signal using an arbitrary waveform generator (AWG) and amplified to drive a dual-polarization IQ modulator. A polarization scrambler is used to introduce RSOP in the fiber link. A variable optical attenuator (VOA) was used to control the optical power. After coherent detection, the received signal was converted to a digital signal using a real-time oscilloscope and processed offline via DSP. To reduce computational complexity while satisfying the requirement of 2 samples per symbol for TED calculation, the received signal was resampled to 55.8 GHz (matching the DSCM signal bandwidth) prior to timing extraction. For the conventional TR schemes, the DSP chain involves frequency offset estimation, subcarrier demultiplexing, and TR, followed by polarization demultiplexing, inter-symbol interference compensation, CPR, and symbol decision for final BER calculation.

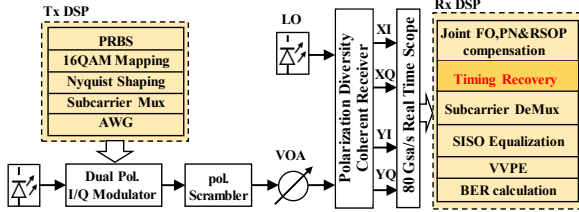


Fig. 4: Experimental setup and the proposed DSP architecture. VOA: variable optical attenuator; LO: local oscillator.

Fig. 5 compares the SFO tracking capability of the investigated schemes under the same loop bandwidth. For the Gardner algorithm, when the SFO is below 120 ppm, the BER increases gradually, which is primarily attributed to the interpolation errors. In contrast, the proposed pilot-aided scheme maintains a significantly lower BER curve. This superior performance stems from the implemented cosine interpolation; as demonstrated in Fig. 3, it achieves a much lower MSE by providing higher timing estimation precision. When the SFO exceeds 120 ppm, the Gardner algorithm suffers from sharp BER degradation and eventually fails. This is because the Gardner algorithm requires a prolonged convergence time that exceeds the finite data length used in the experiment, as demonstrated in Fig. 3. Furthermore, such a large SFO eventually surpasses the tracking range of its feedback loop. Conversely, the proposed pilot-aided

scheme exhibits rapid convergence and maintains stable performance across the entire tested range up to 280 ppm, demonstrating its robust tracking capability.

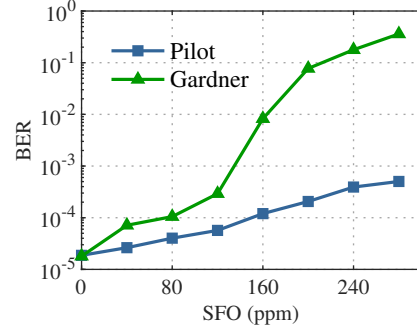


Fig. 5: BER performance versus SFO for the proposed and Gardner schemes.

Fig. 6 illustrates the BER performance of the proposed and conventional schemes as a function of the roll-off factor. Within the roll-off factor range of 0.01 to 0.09, the BER of the proposed scheme remains stable, demonstrating its immunity to roll-off factor variations. In contrast, the Gardner algorithm fails completely when the roll-off factor is reduced to 0.01.

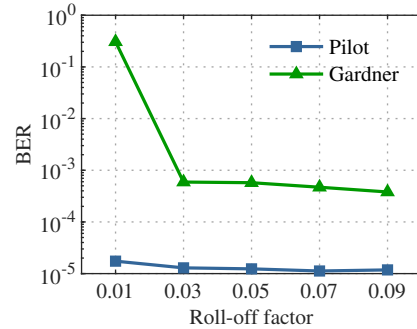


Fig. 6: BER performance versus roll-off factor with 40-ppm SFO.

Conclusions

This paper has proposed a robust pilot-aided TR scheme specifically designed for DSCM systems. By implementing a pilot-division operation, the influence of residual FO and PN on timing extraction is effectively eliminated. Experimental evaluations in a 48-Gbaud PDM-16QAM system demonstrate that the proposed scheme maintains stable operation across an ultra-wide SFO range of up to 280 ppm. In contrast, the Gardner algorithm exhibits a sharp BER performance degradation as the SFO increases from 120 ppm to 160 ppm. Moreover, our scheme ensures high-precision TR even under a low roll-off factor of 0.01. These results confirm that the proposed scheme is a high-performance and robust TR solution, making it a promising candidate for future high-speed coherent interconnects where fast convergence, high precision, and robustness are paramount.

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